

48) If A, B, C , and D are sets, does it follow that $(A \oplus B) \oplus (C \oplus D) = (A \oplus C) \oplus (B \oplus D)$?

Solution: $(A \oplus B) \oplus (C \oplus D) = ((A \oplus B) \oplus C) \oplus D$ (Prob 46)
 $= (A \oplus (B \oplus C)) \oplus D$ (Prob 46)
 $= (A \oplus (C \oplus B)) \oplus D$ (Prob 44(a))
 $= (A \oplus C) \oplus (B \oplus D)$ (Prob 46)
 $= (A \oplus C) \oplus (B \oplus D)$ (Prob 46) $\square$

49) If A, B, C , and D are sets, does it follow that $(A \oplus B) \oplus (C \oplus D) = (A \oplus D) \oplus (B \oplus C)$?

Solution: $(A \oplus B) \oplus (C \oplus D) = ((A \oplus B) \oplus C) \oplus D$ (Prob 46)
 $= (A \oplus (B \oplus C)) \oplus D$ (Prob 46)
 $= D \oplus (A \oplus (B \oplus C))$ (Prob 44(a))
 $= (D \oplus A) \oplus (B \oplus C)$ (Prob 46)
 $= (A \oplus D) \oplus (B \oplus C)$ (Prob 44(a)) $\square$

50) Show that if A and B are finite sets, then $A \cup B$ is a finite set.

Solution: By Defn. 4 from Sec-2.1, Let $|A| = m$ and $|B| = n$, where $m, n \in \mathbb{N}$.

$|A \cup B| = |A| + |B| - |A \cap B|$
 $\leq |A| + |B|$ $\because$ By Defn. 2, $|A \cap B| \geq 0$
 $= m + n$ and by Prob 16(a), $(A \cap B) \subseteq A$

$(m+n) \in \mathbb{N}$, and is finite. $\because |A \cup B| \leq m+n$ and $(A \cap B) \subseteq B$

$A \cup B$ is a finite set $\square$

51) Show that if A is an infinite set, then whenever B is a set, $A \cup B$ is also an infinite set.

Solution: ~~By Defn. 4~~ Suppose for the sake of contradiction, $A \cup B$ is a finite set. By Defn. 4 from Sec-2.1, $|A \cup B| = m$, where $m \in \mathbb{N}$.
 By Prob 16(b), $A \subseteq (A \cup B)$ and $B \subseteq (A \cup B)$. By Defn. of subset, $|A| \leq |A \cup B|$ and $|B| \leq |A \cup B|$. $\therefore |A|, |B| \leq m$. By Defn. 4 from Sec-2.1, Both A and B are finite. $\therefore$ If A is an infinite set, then whenever B is a set, $(A \cup B)$ is also an infinite set. $\square$